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Glass transition and fragility of iron phosphate glasses. II. Effect of mixed-alkali

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Abstract

Changes in the heat capacity from the glass state (C_{pg}) to the supercooled liquid state (C_{pl}), $\Delta C_p = C_{pl} - C_{pg}$, the ratio of heat capacities, C_{pl}/C_{pg}, width of the glass transition temperature, ΔT_g, and activation enthalpy for structural relaxation, ΔH^*, in the glass transition region for the $x Na_2O$–$(20 - x)R_2O$–$32Fe_2O_3$–$48P_2O_5$ (R = K and Cs) mol% glasses, where $x = 0$, 5, 10, 15, 20, have been investigated using differential scanning calorimetry. ΔC_p and C_{pl}/C_{pg} are in the range of 57–68 J/mol/K and 1.41–1.51, respectively, for the $x Na_2O$–$(20 - x)R_2O$–$32Fe_2O_3$–$48P_2O_5$ (R = K and Cs) mol% glasses, which suggests that the alkali-containing iron phosphate glasses, similar to the alkali-free binary iron phosphate glass, also belong in the category of fragile liquids. Adding 20 mol% of alkali oxides to the $40Fe_2O_3$–$60P_2O_5$ (mol%) composition does not change the glass transition behavior or melt fragility very much, but does increase the glass transition temperature, T_g. No mixed-alkali effect is observed on any glass transition properties for the iron phosphate glass.
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1. Introduction

It has been reported that properties related to the structural relaxation, such as shear viscosity [1–3] and glass transition temperature [1,4,5], usually exhibit a negative deviation from additivity when one alkali oxide is replaced by another in homogeneous glasses. Shelby showed [5] that deviations from additivity in bulk thermodynamic properties, viscosity, and glass transition temperature are no greater for mixed-alkali glasses than the deviations typically observed when mixing alkali-free glasses, e.g. SiO_2–GeO_2. Thus, he concluded that deviations from additivity in these properties are due to a 'mixed-glass-former effect' expected whenever two glass-formers are combined.

Moynihan et al. [6] studied the heat capacity and structural relaxation of mixed-alkali $24.4(Na_2O + K_2O)$–$75.6SiO_2$ mol% glasses and found that the heat capacity curves in the glass transition region of all the glasses, for identical heating rates and thermal histories, could be superimposed on the

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same reduced plot. This behavior indicated that the shapes of the structural relaxation functions are the same for all glasses, which reinforced Shelby's viewpoint [5] that there is no unique 'mixed-alkali effect' on thermodynamic or structural relaxation properties and the term should be reserved for properties relating to ionic transport.

On the other hand, however, Komatsu et al. [7,8] reported a mixed-alkali effect for the kinetic fragility and activation enthalpy for structural relaxation for $(20-x)Li_2O$–xNa_2O–$80TeO_2$ mol% glasses. Chryssikos et al. [9] and Chen et al. [10] also indicated the presence of a mixed-alkali effect in the fragility of mixed-alkali metaphosphate glasses.

It has been found in our previous work [11,12] that, unlike most other inorganic oxide glasses, the iron phosphate glasses do not show any mixed-alkali effect for the properties related to ionic motion, such as the dc electrical conductivity, and loss tangent, due to the low alkali ion mobility in these glass. However, the structure relaxation behavior in the glass transition region for alkali-containing iron phosphate glasses has not been reported to our knowledge.

The purpose of the present work was to investigate the glass transition behavior and fragility for several single and mixed-alkali iron phosphate glasses by measuring the change in heat capacity in the glass transition region. The work described herein is a continuation of the work reported in Part II of this series [13], where similar glass transition properties for an alkali-free binary $40Fe_2O_3$–$60P_2O_5$ mol% glass (F40M) were investigated as a function of Fe^{2+} concentration. As mentioned in Part II, iron phosphate glasses, whose compositions are close to that of F40M, are considered promising candidates for vitrifying certain types of nuclear wastes. Many of these nuclear wastes contain one or more alkali oxides. Thus, an understanding of the glass transition behavior of the iron phosphate glasses containing one or more alkali oxides is considered important.

In this paper, the glass transition temperature, heat capacity, activation enthalpy for structural relaxation, and fragility for two series of glasses, whose general compositions are xNa_2O–$(20-x)K_2O$–$32Fe_2O_3$–$48P_2O_5$ (NKFP48) mol%, and xNa_2O–$(20-x)Cs_2O$–$32Fe_2O_3$–$48P_2O_5$ (NCFP48) mol%, with $x = 0$–20, were measured using differential scanning calorimetry. These compositions were derived from the F40M by adding 20 mol% alkali oxides (single or mixed) to the $40Fe_2O_3$–$60P_2O_5$ (mol%) composition, so that the alkali containing glasses had the same Fe/P ratio as that of F40M glass.

2. Experimental procedure

The detailed procedures for glass preparation and heat capacity measurement are given in Part I [13], and the compositions of the as-batched glasses are given in Table 1 along with a sample identification number (ID#) for each glass. The raw materials used to prepare these glasses are Fe_2O_3, P_2O_5, Na_2CO_3, K_2CO_3 or Cs_2CO_3. The last two number after hyphen in the sample ID# (column 1, Table 1) denotes the mol% Na_2O in the glass. The glasses with ID# NKFP48-20 and NCFP48-20 have the same batch composition of $20Na_2O$–$32Fe_2O_3$–$48P_2O_5$ mol%.

It has been reported before [14,15] that the concentration of Fe^{2+} (or Fe^{3+}) in these iron phosphate glasses depends primarily upon the melting temperature, and to some degree, on the melting time. Therefore, all the glasses in the present investigation were prepared by melting the batch at the same temperature (1200 °C) for the same time (1 h), so as to ensure the same Fe^{2+} (and Fe^{3+}) concentration in all these glasses. As expected, measurements by Mössbauer spectroscopy showed that the concentration of Fe^{2+} is nearly the same, $19 \pm 2\%$, in all these glasses. The composition of the glasses as calculated from this measured Fe^{2+} concentration is also included in Table 1. The molecular weight of the glasses, which is required as an input for the calculation of heat capacity, was determined from the calculated glass composition.

3. Results

3.1. Glass transition temperature

An example of the heat capacity vs. temperature (C_p vs. T) curves at different heating rates for

Table 1
Batch and calculated compositions[a] for mixed-alkali iron phosphate glasses

Sample	Batch composition (mol%)				Calculated glass composition (±2.0 mol%)				
	Na_2O	R_2O[b]	Fe_2O_3	P_2O_5	Na_2O	R_2O[b]	FeO	Fe_2O_3	P_2O_5
NKFP48-00	0	20	32	48	0	19.0	10.3	25.2	45.5
NKFP48-05	5	15	32	48	4.8	14.4	8.0	26.7	48.1
NKFP48-10	10	10	32	48	9.6	9.6	8.6	26.3	45.9
NKFP48-15	15	5	32	48	14.4	4.8	8.6	26.3	45.9
NKFP48-20[c]	20	0	32	48	19.0	0	9.7	25.6	45.7
NCFP48-00	0	20	32	48	0	18.8	12.0	24.1	45.1
NCFP48-05	5	15	32	48	4.7	14.1	12.0	24.1	45.1
NCFP48-10	10	10	32	48	9.4	9.4	12.0	24.1	45.1
NCFP48-15	15	5	32	48	13.9	4.6	14.8	22.2	44.5
NCFP48-20[c]	20	0	32	48	19.0	0	9.7	25.6	45.7
F40M	0	0	40	60	0	0	14.1	30.1	55.8

[a] The calculated glass composition for each sample is based on the fraction of Fe^{2+} ions in the glass measured by Mössbauer spectroscopy [13–15].
[b] R = K for the NKFP48 glasses and Cs for the NCFP48 glasses, respectively.
[c] These glasses have the same composition.

the NKFP48-10 glass (for composition see Table 1) is shown in Fig. 1. An increase in the glass transition temperature, T_g, with increasing heating rate is clearly observed. The values of heat capacity for the glass and liquid states, C_{pg} and C_{pl}, respectively, determined from the C_p vs. T curves at 20 K/min are given in Table 2 for all the glasses along with their difference, $\Delta C_p (= C_{pl} - C_{pg})$, and ratio, C_{pl}/C_{pg}.

Fig. 2 shows the plot of T_g obtained from the heat capacity vs. temperature curves with $q_h = 20$ K/min as a function of the $Na_2O/(Na_2O + R_2O)$ (R = K, and Cs) molar ratio for the NKFP48 and NCFP48 glasses. A similar [6] plot for $x Na_2O$–

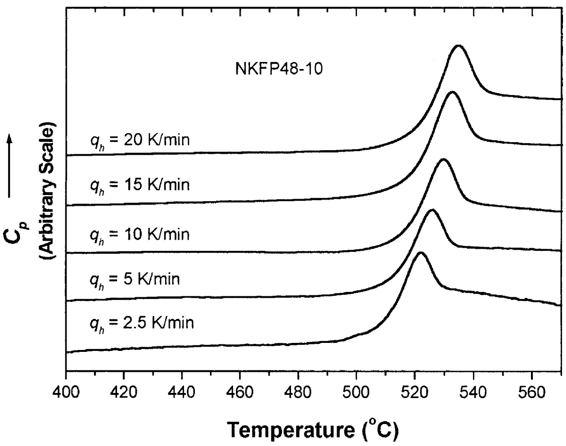

Fig. 1. Heat capacity, C_p, in the glass transition region for the NKFP48-10 glass melted at 1200 °C for 1 h in air. Curves were obtained first by cooling at a rate, q_c, from 550 °C to a temperature well below T_g (~200 °C) and then reheating at a rate, q_h, through the glass transition region with $|q_c/q_h| = 1$. The curves have been intentionally displaced along the vertical direction for clarity.

Table 2
Heat capacity in the glass transition region for the iron phosphate glasses as determined from heat capacity vs. temperature curves

Sample	C_{pg} (±15 J/mol/K)	C_{pl} (±15 J/mol/K)	ΔC_p (±15 J/mol/K)	C_{pl}/C_{pg} (±0.15)
NKFP48-00	138	195	57	1.41
NKFP48-05	140	201	61	1.44
NKFP48-10	143	207	64	1.45
NKFP48-15	147	212	65	1.44
NKFP48-20[a]	152	220	68	1.44
NCFP48-00	124	190	66	1.53
NCFP48-05	132	198	66	1.50
NCFP48-10	135	200	65	1.48
NCFP48-15	142	208	66	1.46
NCFP48-20[a]	152	220	68	1.44
F40M	100	163	63	1.63

C_{pg}, C_{pl}, refers to the heat capacity at the glass transition temperature, T_g and T'_g, respectively, $\Delta C_p = C_{pl} - C_{pg}$ [13].
[a] These glasses have the same composition.

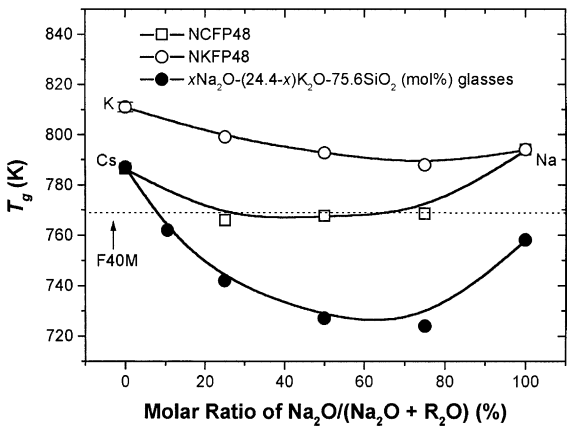

Fig. 2. The glass transition temperature, T_g, measured from heat capacity curves with $q_h = 20$ K/min as a function of the molar ratio of $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) for the NKFP48 (open circles) and NCFP48 (open squares) glasses. The dotted line indicates the T_g for the F40M glass. Data for the $x Na_2O$–$(24.4 - x)K_2O$–$75.6SiO_2$ (mol%) glasses [6] (solid circles) is also given for comparison. Single alkali glasses are marked as using elemental notations. Lines are guide to the eye.

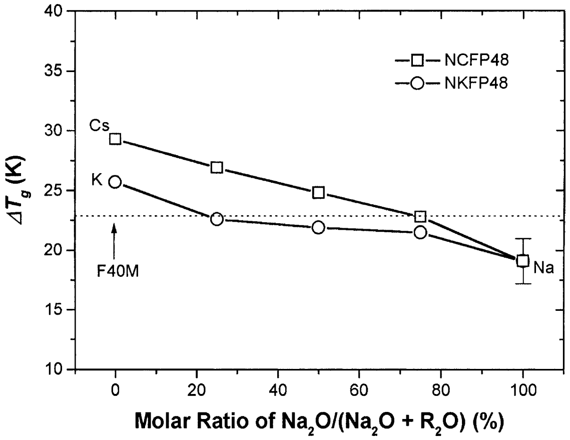

Fig. 3. The width of glass transition temperature, $\Delta T_g = T_g' - T_g$, measured from the heat capacity curves as a function of the molar ratio of $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) for the NKFP48 and NCFP48 glasses. Dotted line is the T_g value for the F40M glass. Single alkali-containing glasses are marked using elemental notations. Lines are guide to the eye.

$(24.4 - x)K_2O$–$75.6SiO_2$ glasses ($q_h = 20$ K/min) is also given in Fig. 2 for comparison. It is to be noted that the fictive temperature T_F is used as the T_g for the mixed-alkali silicate glasses and these two temperatures are shown to be close to each other [6]. The value of T_g (768 K) for the F40M glass obtained from the heat capacity vs. temperature curve with $q_h = 20$ K/min is also shown in Fig. 2 (dotted straight line) for comparison.

A slightly negative deviation in T_g is observed as a function of the alkali oxide ratio for both series (NKFP48 and NCFP48) of mixed-alkali iron phosphate glasses, the deviation being a little larger for the NCFP48 glasses, see Fig. 2. These deviations are similar to that for the mixed-alkali $x Na_2O$–$(24.4 - x)K_2O$–$75.6SiO_2$ glasses [6], but are much smaller. The glass transition temperatures for all of the single alkali iron phosphate glasses ($20Na_2O$–$32Fe_2O_3$–$48P_2O_5$, $20K_2O$–$32Fe_2O_3$–$48P_2O_5$, and $20Cs_2O$–$32Fe_2O_3$–$48P_2O_5$) are higher than that of the alkali-free F40M glass. The single K_2O-containing glass (NKFP48-00) has the highest T_g, while the Cs_2O-containing glass (NCFP48-00) has the lowest T_g for the three single alkali glasses, see Fig. 2.

Fig. 3 shows the glass transition width, $\Delta T_g (= T_g' - T_g)$, at 20 K/min as a function of the $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) molar ratio for the NKFP48 and NCFP48 glasses. No change in ΔT_g was observed when measured at different rates. The value of ΔT_g (23 ± 2 °C) for the alkali-free F40M glass [13] is also shown in Fig. 3 by the dotted straight line. There was no significant difference in ΔT_g for the alkali-free and alkali-containing iron phosphate glasses, but a slight decrease in ΔT_g with increasing soda content is apparent for both glass systems (NKFP48 and NCFP48). For the three single alkali-containing (20 mol%) glasses, ΔT_g decreases in the sequence Cs > K > Na.

3.2. Heat capacity

The change in heat capacity, $\Delta C_p (= C_{pl} - C_{pg})$, was found to be independent of the heating rate used for the measurements. Typical values of ΔC_p at 20 K/min are shown in Fig. 4 as a function of $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) molar ratio for the NKFP48 and NCFP48 glasses. The values of ΔC_p for the $x Na_2O$–$(24.4 - x)K_2O$–$75.6SiO_2$ glasses [6] are also given in Fig. 4 for comparison.

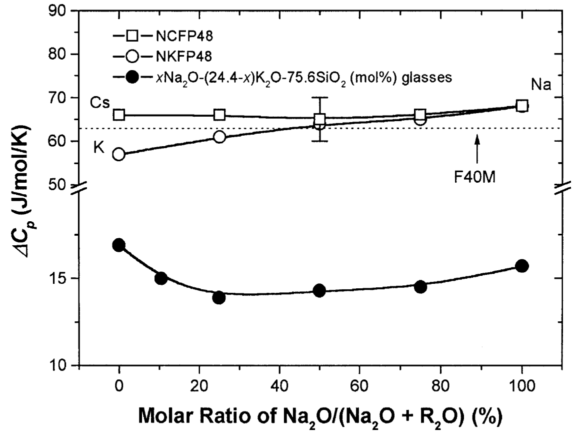

Fig. 4. Heat capacity change from glass to liquid states, $\Delta C_p = C_{pl} - C_{pg}$, as a function of the molar ratio Na_2O/($Na_2O + R_2O$) (R = K and Cs) for the NKFP48 (open circles) and NCFP48 (open squares) glasses. The dotted line indicates the ΔC_p value for the F40M glass. Data for the $x$$Na_2O$–(24.4 − x)K_2O–75.6SiO_2 (mol%) glasses [6] (solid circles) is given for comparison. Single alkali-containing glasses are marked using elemental notations. Lines are guide to the eye.

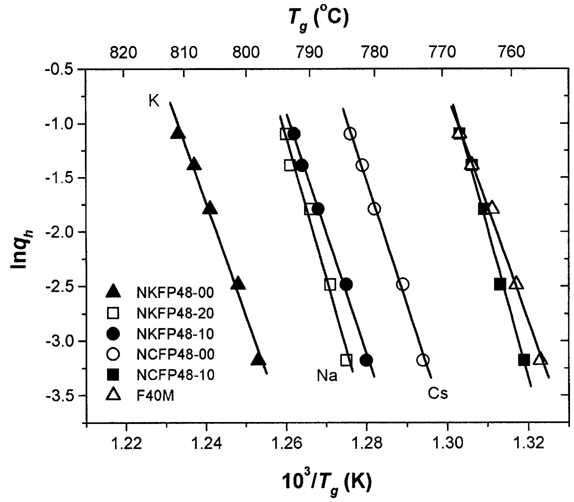

Fig. 5. Plots of $\ln q_h$, vs. $10^3/T_g$ (K) for alkali-free and alkali-containing iron phosphate glasses melted at 1200 °C for 1 h in air. The glass transition temperature, T_g, was determined from heat capacity curves with heating rate, q_h. Single alkali-containing glasses are marked using elemental notations. The solid straight lines are least squares fits to the data points.

A dotted straight line in Fig. 4 indicates the ΔC_p value for the alkali-free F40M glass [13].

As shown in Fig. 4, there is almost no change in ΔC_p for the NCFP48 glasses while ΔC_p slightly increases with increasing soda content for the NKFP48 glasses, also see Table 2. This change in ΔC_p with increasing Na_2O for the NKFP48 glasses is not considered significant, since all the data points fall within the experimental error of ±5 J/mol/K for these measurements. The ΔC_p for the mixed-alkali silicate glasses in Fig. 4 shows a small negative deviation from additivity. However, it was considered [6] that this deviation for ΔC_p was not due to the mixed-alkali effect. Thus, it is clear that there is no mixed-alkali effect in ΔC_p for these iron phosphate glasses.

3.3. Activation enthalpy

The activation enthalpy for structural relaxation in the glass transition region, ΔH^*, was determined using the change in T_g with heating rate, q_h, in the C_p vs. T curves and Eq. (1)

$$d \ln q_h / d(1/T_g) = -\Delta H^*/R, \qquad (1)$$

where R is the ideal gas constant. According to Eq. (1), plots of $\ln q_h$ against $1/T_g$ should be straight lines, as shown in Fig. 5 for the present single and mixed-alkali NKFP48 and NCFP48 iron phosphate glasses. The $\ln q_h$ vs. $1/T_g$ plot for the alkali-free F40M glass is also shown for comparison. A linear correlation with a correlation factor >0.992 is observed in all cases, which suggests that the experimental data are well described by Eq. (1).

The values of ΔH^*, determined from the slope of the straight lines such as shown in Fig. 5 are plotted in Fig. 6 as a function of molar ratio of Na_2O/($Na_2O + R_2O$) (R = K and Cs) for the NKFP48 and NCFP48 glasses. Fig. 6 also includes similar ΔH^* values for the $x$$Na_2O$–(20 − x)Li_2O–80TeO_2 (mol%) glasses [8]. The dotted straight line in Fig. 6 indicates the ΔH^* value (862 ± 35 kJ/mol) for the alkali-free F40M glass [13].

As shown in Fig. 6, the ΔH^* for the alkali-containing iron phosphate glasses are a little larger than that for the alkali-free F40M glass. It is also observed (Fig. 6) that, unlike the ΔH^* for the mixed-alkali tellurite glasses, the ΔH^* for the mixed-alkali iron phosphate glasses do not show a negative deviation from additivity with additions

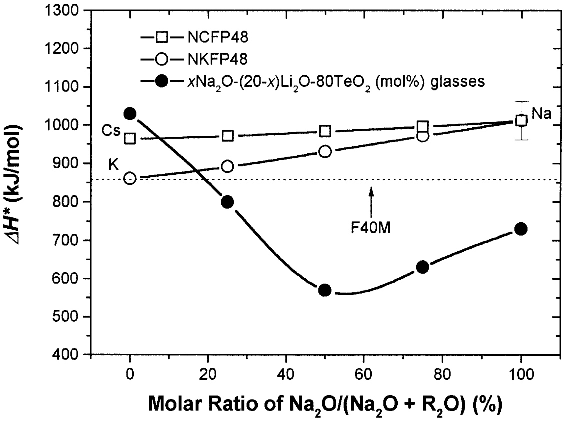

Fig. 6. The activation enthalpy, ΔH^*, for structural relaxation as a function of the molar ratio $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) for the NKFP48 (open circles) and NCFP48 (open squares) glasses. Dotted line is the ΔH^* value for the F40M glass. Data for $x Na_2O$–$(20 - x)Li_2O$–$80TeO_2$ (mol%) glasses [8] (solid circles) is given for comparison. Single alkali-containing glasses are marked using elemental notations. Lines are guide to the eye.

of a second alkali. A small increase in ΔH^* with increasing Na_2O content occurred in both of the NKFP48 and NCFP48 glasses, but no maxima or minima is observed in these curves. Obviously, there is no mixed-alkali effect in the activation enthalpy for structural relaxation for the iron phosphate glasses.

4. Discussion

4.1. Mixed-alkali effect

The mixed-alkali effect is a phenomenon observed in many oxide glasses in which several ionic transport-related properties change as a function of composition in a highly non-additive fashion when one alkali oxide is replaced by another. Mixed-alkali iron phosphate glasses seem to be as exception. As reported previously [11,12], the alkali ions in iron phosphate glasses are believe to have a very low mobility, so that the mixed-alkali effect is absent in properties such as the dc electrical resistivity or dielectric loss for these glasses.

It has been suggested [5,6] that the properties related to structural relaxation in a glass should not respond to the typical mixed-alkali-type effect. The current results are in agreement with this viewpoint. For example, neither ΔC_p (Fig. 4) nor ΔH^* (Fig. 6) for any of these NKFP48 and NCFP48 glasses show a pronounced maxima or minima when plotted as a function of alkali ratio.

In Fig. 2, T_g for both types of mixed-alkali iron phosphate glasses (NKFP48 and NCFP48) shows a slightly negative deviation (minima) from additivity, but this deviation is considerably smaller than that observed in the T_g for the mixed-alkali silicate glasses ($x Na_2O$–$(24.4 - x)K_2O$–$75.6SiO_2$). Furthermore, none of the other properties for these iron phosphate glasses such as the chemical durability [11,12], dc electrical conductivity or resistivity [11,12], dielectric constant [11,12] show any kind of mixed-alkali effect. Thus, it is highly unlikely that the small minima present in the T_g (as a function of alkali ratio) for these iron phosphate glasses occurred due to the mixed-alkali effect. This conclusion is also in agreement with Shelby's general comment [5] that there is no unique mixed-alkali effect for glass transition properties and that the term should be reserved for properties relating to ionic transport. However, opinions different from Shelby's hypothesis that there should be a mixed-alkali effect for properties related to glass transition temperature or activation enthalpy for structural relaxation are also available in the literature [8–10].

4.2. Fragility

The ratio of C_{pl}/C_{pg} was used as a parameter by Angell [16,17] to classify strong-fragile liquids and according to him the liquids having a $C_{pl}/C_{pg} > 1.1$ are of fragile types. The values of C_{pl}/C_{pg} for the present alkali-containing iron phosphate glasses (NKFP48 and NCFP48) range between 1.41 and 1.53 (Table 2), which put these glasses into the fragile category. As shown in Table 2, the alkali-free F40M glass ($C_{pl}/C_{pg} \sim 1.63$) also belongs in the fragile category. No significant difference or systematic change in C_{pl}/C_{pg} for the alkali-containing glasses as a function of alkali ratio is observed, which suggests that the mixed-alkali effect is either absent in these glasses or not observable in this particular property of C_{pl}/C_{pg}.

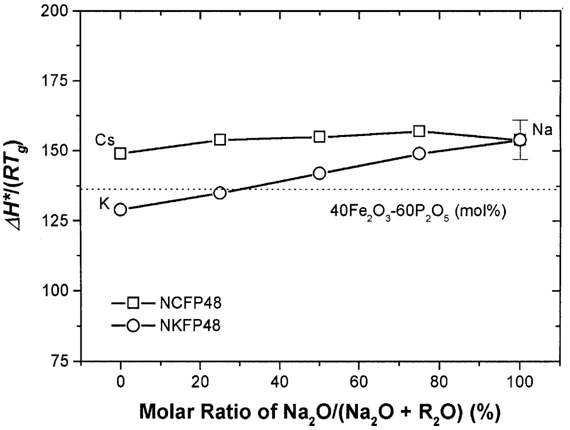

Fig. 7. The fragility parameter, $\Delta H^*/(RT_g)$, as a function of the molar ratio $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) for NKFP48 (open circles) and NCFP48 (open squares) glasses. The dotted line is the fragility parameter for the F40M glass. Single alkali-containing glasses are marked using elemental notations. Lines are guide to the eye.

As mentioned in Part II of this paper, parameter of $[-T_g(\mathrm{d}\ln\eta/\mathrm{d}T)|_{T_g}](= \Delta H^*/(RT_g))$ introduced by Brüning and Sutton [18] also can be used to evaluate the fragility of melts. Fig. 7 shows plots of $\Delta H^*/(RT_g)$ as a function of the $Na_2O/(Na_2O + R_2O)$ (R = K and Cs) molar ratio for the NKFP48 (open squares) and NCFP48 (open circles) glasses. The fragility value of 136 for the F40M glass [13] is also indicated in Fig. 7 by the dotted line. The fragility for the alkali-containing glass is close to that for the alkali-free glass, indicating that the fragility for the alkali-free, and single and mixed-alkali iron phosphate glasses, up to 20 mol% alkali oxides, is about the same.

Previous investigation by Mössbauer, Raman and infrared spectroscopy indicated [11,12] that adding alkali oxide, at least up to 20 mol%, to the binary F40M composition did not significantly change the glass structure, i.e. did not change the concentration of Fe–O–P and P–O–P bonds and coordination environment of different ions in the glasses. Since the fragility of a glass primarily depends on the glass structure, it is expected that the fragility of these alkali-free, and alkali-containing iron phosphate glasses would be the same.

It is interesting to note that all the Na_2O and Cs_2O containing glasses (single or mixed) have nearly a constant fragility parameter (dimensionless) of, $\sim 154 \pm 4$ (Fig. 7), which is larger than that of the alkali-free F40M glass ($\sim$136) or single alkali K_2O iron phosphate glass ($\sim$129). In the mixed-alkali K_2O–Na_2O iron phosphate glasses (NKFP48), the fragility parameter increases with increasing Na_2O. In other words, the fragility for all the single and mixed-alkali Na_2O–Cs_2O iron phosphate glasses is the same and they are more fragile than the K_2O-containing (single alkali) iron phosphate glass. The fragility for the mixed K_2O–Na_2O iron phosphate glasses increases with increasing Na_2O.

The electric field strength (z/r^2) for these three alkali ions increases as $Cs^+ < K^+ < Na^+$. Thus, if the bond strength between the alkali and oxygen ions is the only determining factor for fragility, then the fragility parameter for the single alkali containing iron phosphate glasses is expected to decrease in the sequence $Cs_2O > K_2O > Na_2O$, much like the way shown in Fig. 3 for the glass transition width, ΔT_g. The results in Fig. 7 are clearly different from the trend mentioned above. It appears, therefore, that several factors other than the bond strength may be responsible for the fragility of a glass forming liquid. In any case, the fragility parameter for all the alkali iron phosphate glasses investigated in the present work only ranges from 129 to 157, which is not considered to be a very large difference. Therefore, the degree of fragility for all these iron phosphate glasses is approximately the same, an inference also drawn from by comparing the values for C_{pl}/C_{pg} for these glasses.

No maxima or minima appear in the plots of fragility parameter vs. the molar ratio for these mixed-alkali iron phosphate glasses, Fig. 7, which, again, suggests that the mixed-alkali effect is not observable in the fragility for these glasses.

4.3. Glass transition temperature

Commonly, it is expected that adding alkali oxides, which act as glass network modifiers in oxide glasses will reduce the viscosity and the melting temperature. As a result, the glass transition temperature is expected to decrease. However, the T_g values for all the NKFP48 and NCFP48 glasses in the present investigation are larger than, or, in some cases, equal to that for the F40M glass (768 ± 2 K), see Fig. 2. These results suggest that the alkali ion environment in these iron phosphate glasses may be different from that in other oxide glasses.

It was found previously [11,12] that the alkali ions in the iron phosphate glasses tend to form $RFe[P_2O_7]$ units (R = Na, K and Cs), which reduced mobility of the alkali ions in the glass. In the structure of the NKFP48 and NCFP48 glasses, pyrophosphate $(P_2O_7)^{4-}$ units are bonded to oxygen polyhedra that contain the Fe^{2+} and Fe^{3+} ions. In $(P_2O_7)^{4-}$ units, two PO_4 tetrahedra, composed of one bridging and three non-bridging oxygens, are joined by a P–O–P bond.

When alkali oxide (R_2O) is added, the R^+ ions can interact with the negatively charged non-bridging $P–O^-$ oxygen ions to form $RFe[P_2O_7]$ units. Due to the existence of the $RFe[P_2O_7]$ units, the bonding connection between the pyrophosphate units and iron oxygen polyhedra is improved. Thus, the structural relaxation takes place at a higher temperature for these alkali-containing glasses as compared to the alkali-free F40M iron phosphate glasses.

5. Conclusions

None of the glass transition properties, such as the width of the glass transition region (ΔT_g), change in heat capacity (ΔC_p), heat capacity ratio (C_{pl}/C_{pg}), activation enthalpy for structural relaxation (ΔH^*), and the fragility parameter ($\Delta H^*/(RT_g)$) for the iron phosphate glasses in the present investigation change significantly when a single or mixed-alkali oxides is added, up to 20 mol%, to the binary $40Fe_2O_3$–$60P_2O_5$ (mol%) composition. Also, no mixed-alkali effect is observed on any of these properties for these iron phosphate glasses.

The glass transition temperature slightly increases for the alkali iron phosphate glasses, as compared to the alkali-free iron phosphate glass, presumably due to the formation of $RFe[P_2O_7]$ groups in the structure, that improved the bonding connections between the pyrophosphate units and the iron oxygen polyhedra in the glass.

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